

# Dimitris Tsitsigkos

Software Engineer (PhD)

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## Work Experience

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### Postdoctoral researcher (Data Science and Engineering team)

Feb 2025 – Present

#### Archimedes Research on AI, Data Science and Algorithms, Athens, Greece

- Designed and implemented high-performance, in-memory multi-dimensional and vector indices in C++, using parallelism and hardware-aware optimizations to improve query throughput and latency
- Technologies: C++, OpenMP, SIMD

### PhD fellow

Jan 2024 – Dec 2024

#### Department of Computer Science & Engineering, Univ. of Ioannina, Greece

- Contributed to the implementation of spatial index structures in C++ for distance queries (k-NN, range, distance joins), improving query performance compared to existing approaches
- Contributed to the design and development of a prototype distributed spatial data management system using MPI and OpenMP, reducing memory consumption by over 55% while improving query performance.
- Technologies: C++, OpenMP, MPI

### PhD fellow

Jul 2019 – Dec 2024

#### Information Management Systems Institute, Athena Research Center, Athens, Greece

- PhD Research:
  - Optimized B+-Tree Index Structures
    - \* Implemented BS-tree, a gapped data-parallel B+-tree, and its compressed variant CBS-tree for high-performance in-memory indexing
    - \* Both BS-tree and CBS-tree achieved  $1.5\times$ – $2.5\times$  higher throughput versus B+-trees, learned indexes, and trie-based approaches across different read and update workloads
    - \* CBS-tree uses 56%–94% less memory than competing indexes on compressible datasets while maintaining superior throughput
    - \* Designed SIMD-friendly node layouts, branchless search, and efficient gap handling to fully exploit data-parallel CPU features, boosting single-threaded performance and scaling across cores
  - Indexing and Partitioning for Non-Point Spatial Data
    - \* Developed two-layer space-oriented partitioning for indexing non-point spatial objects
    - \* Reduced spatial join cost by  $\sim 50\%$  compared to existing spatial indexing methods
    - \* Achieved  $2\text{--}3\times$  faster range query performance vs R-trees, quad-trees and grid-based indexes
    - \* Designed techniques supporting parallel spatial query processing
  - Parallel Spatial Join Processing
    - \* Implemented a parallel spatial join algorithm for rectangles using the Partition-Based Spatial Merge (PBSM)
    - \* Optimized 1D/2D partitioning, sweep strategies, and data-driven parameter tuning to maximize join performance
    - \* Achieved parallel spatial joins of tens of millions of rectangles in  $<1\text{ s}$  on multi-core systems, demonstrating high scalability and efficiency
- Participated in European funding project:
  - MORE: Management of Real-time Energy data.
    - \* Built parallel continuous evaluation module in Java for sliding-window aggregations, enabling low-latency energy analytics
    - \* Implemented parallel versions of time-series analytics algorithms in Python using Dask to improve efficiency and scalability for large-scale data processing
- Technologies: C++, OpenMP, SIMD, Python, Dask

### Software Engineer

Apr 2017 – Nov 2018

#### Hellenic Army Information Technology Support Center, Athens, Greece

- Maintained and extended Java backend applications for internal systems, improving performance and implementing new features
- Technologies: Java, Oracle Database, JSF.

### Research Engineer

Dec 2012 – Jun 2019

#### Information Management Systems Institute, Athena Research Center, Athens, Greece

Participated in several Greek and European funding projects, including:

- Amnesia: A platform for anonymizing relational, multi-dimensional, and hierarchical data.
  - Implemented an open-source data anonymization platform for in-memory anonymization, supporting both web and desktop applications
  - Designed the architecture and full stack, with a Java backend providing RESTful APIs (Spring) and a web frontend using JavaScript and HTML
  - Integrated the platform into the OpenAIRE ecosystem, enabling global researchers, data providers, and companies to anonymize data efficiently
- MoDisSENSE: A Distributed Spatio-Temporal and Textual Processing Platform for Social.
  - Built distributed spatio-temporal processing (Java/Hadoop/HBase) for trajectory reconstruction and POI analytics with RESTful APIs and PostgreSQL
- Technologies: Java, RESTful Web Services, Spring, MapReduce, Hadoop, HBase, PostgreSQL

## Technical Skills

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|---------------------------------|------------------------------------------------------------------------------------|
| Languages:                      | C++, Java, Python                                                                  |
| Data Management:                | MySQL, PostgreSQL, PostGIS, HBase                                                  |
| Parallel & Distributed Systems: | OpenMP, SIMD, Spark, MPI, Hadoop, Dask                                             |
| Backend & Web:                  | Spring, RESTful Web Services, JavaScript, JSP, JSF                                 |
| Operating Systems:              | Ubuntu, Microsoft Windows, macOS                                                   |
| Other:                          | Data management, in-memory indexing & optimization, parallel & distributed systems |

## Education

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### PhD, Data Management

Jul 2019 – Dec 2024

Department of Computer Science & Engineering, University of Ioannina (Uoi) in collaboration with Athena Research Center

- Visiting Researcher (PhD Collaboration), Institute of Computer Science (Data Management research group), Johannes Gutenberg University Mainz, Germany (May 2022 – Jul 2022)

*Thesis: In-memory Indexing for Parallel Processing of Single and Multi-Dimensional Queries.*

Published papers in: ICDE, VLDBJ, TKDE and SIG-SPATIAL

### MSc, Computing Systems: Software and Hardware, Computer Science

Nov 2012 – Sep 2016

Department of Informatics and Telecommunications, National and Kapodistrian University of Athens (NKUA)

*Thesis: Complex Event Processing (CEP) for Intrusion Detection.*

### BSc, Computer Science

Sep 2006 – Jun 2012

Department of Informatics and Telecommunications, National and Kapodistrian University of Athens (NKUA)

*Thesis: Clustering Wikipedia resources.*

## Additional Experience & Awards

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- [Best Research Paper Runner-Up](#) — ICDE 2026
- Published papers in top-tier peer-reviewed database conferences and journals (h-index: 5), including SIGMOD, VLDBJ, ICDE and EDBT
- 3rd place — [Future of Database Programming Contest](#), Athens (Mar 2025)
- Program Committee member and external reviewer for 8+ international conferences and journals in data management. Distinguished PC Member, ICDE 2026
- Volunteer: European Data Forum 2014, EDBT/ICDT 2023 Joint Conference, 6th ACM Europe Summer School on Data Science 2025, HDMS 2025

## External Links

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- LinkedIn: [https://gr.linkedin.com/in/dimitris\\_tsitsigkos](https://gr.linkedin.com/in/dimitris_tsitsigkos)
- Google Scholar: [https://scholar.google.com/citations?user=z\\_KVU0QAAAAJ](https://scholar.google.com/citations?user=z_KVU0QAAAAJ)
- DBLP: <https://dblp.org/pid/163/0537.html>
- GitHub: <https://github.com/dTsitsigkos>
- Personal website: <https://dtsitsigkos.github.io/>

## Languages

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- Greek (Native)
- English (Fluent)